Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 1

Part 1. Give the definitions

1. Strict local minimum.
2. Positive definite and positive semidefinite matrix.
3. L-smoothness (Lipschitz gradient).
4. Lagrangian.
5. Karush–Kuhn–Tucker (KKT) conditions.
6. Dual function and dual problem.

Part 2. State and prove

1. The negative gradient is the direction of steepest local decrease (among $\|d\|_2 = 1$).
2. Guaranteed decrease of a GD step for L-smooth f: $f(x^+) \le f(x) - \alpha(1 - \frac{L\alpha}{2})\|\nabla f(x)\|^2$.

Part 3. Problems

1. Find the stationary points of $f(x, y) = 3x^2 - 2xy + 3y^2 - 4x - 4y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = x^2 - xy + y^2$, perform three steps of gradient descent from $x_0 = (6,\ 0)$ with step $\alpha = \frac{1}{3}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 2

Part 1. Give the definitions

1. Optimization problem: decision variable, objective, feasible set.
2. Local minimum.
3. Global minimum.
4. Gradient (via the linear approximation).
5. Stationary point and saddle point.
6. Convex function.

Part 2. State and prove

1. First-order necessary condition: at an unconstrained local minimizer $\nabla f(x^\star) = 0$.
2. Stability of GD on a quadratic $\frac{1}{2}x^\top Qx - b^\top x$, $Q \succ 0$: convergence from every $x_0 \iff 0 < \alpha < 2/\lambda_{\max}(Q)$.

Part 3. Problems

1. Find the stationary points of $f(x, y) = 5x^2 + 6xy + 5y^2 + 10x + 6y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = x^2 - xy + y^2$, perform three steps of gradient descent from $x_0 = (8,\ 4)$ with step $\alpha = \frac{1}{2}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 3

Part 1. Give the definitions

1. Directional derivative.
2. Hessian.
3. Feasible direction.
4. Convex combination and convex set.
5. Strictly convex and μ-strongly convex function.
6. Condition number.

Part 2. State and prove

1. Second-order necessary condition: at an unconstrained local minimizer $\nabla^2 f(x^\star) \succeq 0$.
2. Strong convexity implies the Polyak–Lojasiewicz condition.

Part 3. Problems

1. Find the stationary points of $f(x, y) = 5x^2 + 2xy + 5y^2 - 30x - 6y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = 2(x-1)^2 + (y+1)^2$, perform three steps of gradient descent from $x_0 = (10,\ 8)$ with step $\alpha = \frac{1}{3}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 4

Part 1. Give the definitions

1. Polyak–Lojasiewicz condition.
2. Descent direction.
3. Gradient descent with a constant step.
4. Armijo condition (backtracking).
5. Active constraint.
6. Newton's method (Newton direction).

Part 2. State and prove

1. Weak duality: $q(\lambda, \nu) \le f(x)$ for every feasible x and $\lambda \ge 0$.
2. Linear convergence of GD under PL: with $\alpha = 1/L$, $f(x_k) - f^\star \le (1 - \mu/L)^k (f(x_0) - f^\star)$.

Part 3. Problems

1. Find the stationary points of $f(x, y) = 5x^2 + 4xy + 2y^2 - 10x - 4y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = (x-1)^2 + (y+2)^2$, perform three steps of gradient descent from $x_0 = (9,\ 6)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 5

Part 1. Give the definitions

1. Gradient (via the linear approximation).
2. Convex function.
3. Condition number.
4. Gradient descent with a constant step.
5. Armijo condition (backtracking).
6. Karush–Kuhn–Tucker (KKT) conditions.

Part 2. State and prove

1. A symmetric $H \succeq 0$ if and only if all its eigenvalues are nonnegative.
2. For $f(x) = \frac{1}{2}x^\top Ax - b^\top x + c$, $A = A^\top \succ 0$: $\nabla f = Ax - b$, $\nabla^2 f = A$, and $x^\star = A^{-1}b$ is the unique global minimizer.

Part 3. Problems

1. Find the stationary points of $f(x, y) = 3x^2 + 2xy + 3y^2 - 10x - 14y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = 3x^2 + (y - 1)^2$, perform three steps of gradient descent from $x_0 = (4,\ 9)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 6

Part 1. Give the definitions

1. Strict local minimum.
2. Positive definite and positive semidefinite matrix.
3. Feasible direction.
4. Convex combination and convex set.
5. L-smoothness (Lipschitz gradient).
6. Polyak–Lojasiewicz condition.

Part 2. State and prove

1. The intersection of any family of convex sets is convex.
2. Newton's method reaches the minimizer of a quadratic with $Q \succ 0$ in one step.

Part 3. Problems

1. Find the stationary points of $f(x, y) = 3x^2 + 4xy + 3y^2 + 8x + 12y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = (x-2)^2 + 3(y+1)^2$, perform three steps of gradient descent from $x_0 = (10,\ 3)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 7

Part 1. Give the definitions

1. Directional derivative.
2. Hessian.
3. Stationary point and saddle point.
4. Strictly convex and μ-strongly convex function.
5. Active constraint.
6. Dual function and dual problem.

Part 2. State and prove

1. For $f(x) = \frac{1}{2}x^\top Ax - b^\top x + c$, $A = A^\top \succ 0$: $\nabla f = Ax - b$, $\nabla^2 f = A$, and $x^\star = A^{-1}b$ is the unique global minimizer.
2. The negative gradient is the direction of steepest local decrease (among $\|d\|_2 = 1$).

Part 3. Problems

1. Find the stationary points of $f(x, y) = 3x^2 + 4xy + 6y^2 + 8x - 4y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = 2x^2 + 2y^2$, perform three steps of gradient descent from $x_0 = (6,\ -4)$ with step $\alpha = \frac{1}{8}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 8

Part 1. Give the definitions

1. Optimization problem: decision variable, objective, feasible set.
2. Local minimum.
3. Global minimum.
4. Descent direction.
5. Lagrangian.
6. Newton's method (Newton direction).

Part 2. State and prove

1. Second-order necessary condition: at an unconstrained local minimizer $\nabla^2 f(x^\star) \succeq 0$.
2. Linear convergence of GD under PL: with $\alpha = 1/L$, $f(x_k) - f^\star \le (1 - \mu/L)^k (f(x_0) - f^\star)$.

Part 3. Problems

1. Find the stationary points of $f(x, y) = 4x^2 - 4xy + 4y^2 - 16x + 20y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = 2(x-2)^2 + (y-1)^2$, perform three steps of gradient descent from $x_0 = (1,\ 9)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 9

Part 1. Give the definitions

1. Strict local minimum.
2. Directional derivative.
3. Strictly convex and μ-strongly convex function.
4. Condition number.
5. Armijo condition (backtracking).
6. Dual function and dual problem.

Part 2. State and prove

1. Strong convexity implies the Polyak–Lojasiewicz condition.
2. Stability of GD on a quadratic $\frac{1}{2}x^\top Qx - b^\top x$, $Q \succ 0$: convergence from every $x_0 \iff 0 < \alpha < 2/\lambda_{\max}(Q)$.

Part 3. Problems

1. Find the stationary points of $f(x, y) = 2x^2 - xy + 2y^2 - 9x + 6y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = x^2 - xy + y^2$, perform three steps of gradient descent from $x_0 = (4,\ 0)$ with step $\alpha = \frac{1}{2}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 10

Part 1. Give the definitions

1. Local minimum.
2. Hessian.
3. Feasible direction.
4. Convex function.
5. Descent direction.
6. Newton's method (Newton direction).

Part 2. State and prove

1. A symmetric $H \succeq 0$ if and only if all its eigenvalues are nonnegative.
2. Guaranteed decrease of a GD step for L-smooth f: $f(x^+) \le f(x) - \alpha(1 - \frac{L\alpha}{2})\|\nabla f(x)\|^2$.

Part 3. Problems

1. Find the stationary points of $f(x, y) = 3x^2 + 3xy + 3y^2 + 9x + 9y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = (x-1)^2 + 2(y+1)^2$, perform three steps of gradient descent from $x_0 = (5,\ 3)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 11

Part 1. Give the definitions

1. Optimization problem: decision variable, objective, feasible set.
2. Gradient (via the linear approximation).
3. Stationary point and saddle point.
4. Convex combination and convex set.
5. Lagrangian.
6. Active constraint.

Part 2. State and prove

1. First-order necessary condition: at an unconstrained local minimizer $\nabla f(x^\star) = 0$.
2. Weak duality: $q(\lambda, \nu) \le f(x)$ for every feasible x and $\lambda \ge 0$.

Part 3. Problems

1. Find the stationary points of $f(x, y) = x^2 + xy + y^2 - x + y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = 4(x-1)^2 + (y-1)^2$, perform three steps of gradient descent from $x_0 = (3,\ 17)$ with step $\alpha = \frac{1}{8}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 12

Part 1. Give the definitions

1. Global minimum.
2. Positive definite and positive semidefinite matrix.
3. L-smoothness (Lipschitz gradient).
4. Polyak–Lojasiewicz condition.
5. Gradient descent with a constant step.
6. Karush–Kuhn–Tucker (KKT) conditions.

Part 2. State and prove

1. The intersection of any family of convex sets is convex.
2. Newton's method reaches the minimizer of a quadratic with $Q \succ 0$ in one step.

Part 3. Problems

1. Find the stationary points of $f(x, y) = 2x^2 + 4xy + 5y^2 - 4x - 16y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = 3(x - 1)^2 + y^2$, perform three steps of gradient descent from $x_0 = (5,\ -8)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 13

Part 1. Give the definitions

1. Stationary point and saddle point.
2. Positive definite and positive semidefinite matrix.
3. Feasible direction.
4. Descent direction.
5. Gradient descent with a constant step.
6. Karush–Kuhn–Tucker (KKT) conditions.

Part 2. State and prove

1. First-order necessary condition: at an unconstrained local minimizer $\nabla f(x^\star) = 0$.
2. Stability of GD on a quadratic $\frac{1}{2}x^\top Qx - b^\top x$, $Q \succ 0$: convergence from every $x_0 \iff 0 < \alpha < 2/\lambda_{\max}(Q)$.

Part 3. Problems

1. Find the stationary points of $f(x, y) = x^2 + 3y^2 - 2x + 18y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = x^2 + xy + y^2$, perform three steps of gradient descent from $x_0 = (3,\ 0)$ with step $\alpha = \frac{1}{3}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 14

Part 1. Give the definitions

1. Optimization problem: decision variable, objective, feasible set.
2. Strict local minimum.
3. Directional derivative.
4. Convex function.
5. Polyak–Lojasiewicz condition.
6. Active constraint.

Part 2. State and prove

1. For $f(x) = \frac{1}{2}x^\top Ax - b^\top x + c$, $A = A^\top \succ 0$: $\nabla f = Ax - b$, $\nabla^2 f = A$, and $x^\star = A^{-1}b$ is the unique global minimizer.
2. Newton's method reaches the minimizer of a quadratic with $Q \succ 0$ in one step.

Part 3. Problems

1. Find the stationary points of $f(x, y) = 5x^2 - 4xy + 2y^2 - 16x + 4y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = x^2 + 4y^2$, perform three steps of gradient descent from $x_0 = (16,\ 1)$ with step $\alpha = \frac{1}{8}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 15

Part 1. Give the definitions

1. Global minimum.
2. Gradient (via the linear approximation).
3. Convex combination and convex set.
4. L-smoothness (Lipschitz gradient).
5. Armijo condition (backtracking).
6. Dual function and dual problem.

Part 2. State and prove

1. Strong convexity implies the Polyak–Lojasiewicz condition.
2. Linear convergence of GD under PL: with $\alpha = 1/L$, $f(x_k) - f^\star \le (1 - \mu/L)^k (f(x_0) - f^\star)$.

Part 3. Problems

1. Find the stationary points of $f(x, y) = 6x^2 + 4xy + 3y^2 - 16x - 10y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = x^2 + xy + y^2$, perform three steps of gradient descent from $x_0 = (0,\ 8)$ with step $\alpha = \frac{1}{2}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 16

Part 1. Give the definitions

1. Local minimum.
2. Hessian.
3. Strictly convex and μ-strongly convex function.
4. Condition number.
5. Lagrangian.
6. Newton's method (Newton direction).

Part 2. State and prove

1. A symmetric $H \succeq 0$ if and only if all its eigenvalues are nonnegative.
2. Guaranteed decrease of a GD step for L-smooth f: $f(x^+) \leq f(x) - \alpha(1 - \frac{L\alpha}{2})\|\nabla f(x)\|^2$.

Part 3. Problems

1. Find the stationary points of $f(x, y) = x^2 - xy + y^2 + x - 2y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = (x + 1)^2 + 2y^2$, perform three steps of gradient descent from $x_0 = (7,\ 2)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 17

Part 1. Give the definitions

1. Gradient (via the linear approximation).
2. Stationary point and saddle point.
3. Positive definite and positive semidefinite matrix.
4. L-smoothness (Lipschitz gradient).
5. Lagrangian.
6. Newton's method (Newton direction).

Part 2. State and prove

1. Weak duality: $q(\lambda, \nu) \le f(x)$ for every feasible x and $\lambda \ge 0$.
2. The negative gradient is the direction of steepest local decrease (among $\|d\|_2 = 1$).

Part 3. Problems

1. Find the stationary points of $f(x, y) = 2x^2 + 2xy + 2y^2 - 8x - 4y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = 2x^2 + (y - 2)^2$, perform three steps of gradient descent from $x_0 = (4,\ 6)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 18

Part 1. Give the definitions

1. Strict local minimum.
2. Feasible direction.
3. Convex combination and convex set.
4. Descent direction.
5. Karush–Kuhn–Tucker (KKT) conditions.
6. Dual function and dual problem.

Part 2. State and prove

1. Second-order necessary condition: at an unconstrained local minimizer $\nabla^2 f(x^\star) \succeq 0$.
2. The intersection of any family of convex sets is convex.

Part 3. Problems

1. Find the stationary points of $f(x, y) = 2x^2 - 2xy + 2y^2 + 4x - 8y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = x^2 + xy + y^2$, perform three steps of gradient descent from $x_0 = (4,\ 0)$ with step $\alpha = \frac{1}{2}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 19

Part 1. Give the definitions

1. Global minimum.
2. Hessian.
3. Convex function.
4. Strictly convex and μ-strongly convex function.
5. Armijo condition (backtracking).
6. Active constraint.

Part 2. State and prove

1. First-order necessary condition: at an unconstrained local minimizer $\nabla f(x^\star) = 0$.
2. The negative gradient is the direction of steepest local decrease (among $\|d\|_2 = 1$).

Part 3. Problems

1. Find the stationary points of $f(x, y) = 4x^2 + 2xy + 4y^2 - 20x - 20y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = x^2 + 3y^2$, perform three steps of gradient descent from $x_0 = (8,\ 4)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 20

Part 1. Give the definitions

1. Optimization problem: decision variable, objective, feasible set.
2. Local minimum.
3. Directional derivative.
4. Polyak–Lojasiewicz condition.
5. Condition number.
6. Gradient descent with a constant step.

Part 2. State and prove

1. For $f(x) = \frac{1}{2}x^\top Ax - b^\top x + c$, $A = A^\top \succ 0$: $\nabla f = Ax - b$, $\nabla^2 f = A$, and $x^\star = A^{-1}b$ is the unique global minimizer.
2. Guaranteed decrease of a GD step for L-smooth f: $f(x^+) \le f(x) - \alpha(1 - \frac{L\alpha}{2})\|\nabla f(x)\|^2$.

Part 3. Problems

1. Find the stationary points of $f(x, y) = 2x^2 + xy + 2y^2 + 3x - 3y$ and determine their type. Explain why your conclusion holds. *Follow-up:* what are μ and L for this function?
2. For $f(x, y) = x^2 + 2y^2$, perform three steps of gradient descent from $x_0 = (9,\ 9)$ with step $\alpha = \frac{1}{3}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.